

The Route-Capturing Game

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Abstract

In this paper, we are concerned with a strategic game called the route-capturing game, in which the players compete for a set of given resources with finite budgets. The game is modeled with a graph, where the vertices stands for the players, and edges stands for the resources to be captured. Given that each player has the same budget and can spends a bounded amount of budget to capture each edge, properties of the Nash equilibria of some classes of the game are investigated. Experimental results are also shown to characterize the constitution of the Nash equilibria.

1 Introduction

The situation of competing for resources with limited budgets arises enormously in society. This category of problems are normally formulated as a kind of *resource allocation problems* [2, 9, 16], and are widely investigated in the fields of economics [10, 12, 13] and telecommunication [8, 11, 18]. In game-theoretic researches, related problems such as *congestion game* [1, 3, 4, 5, 7, 15, 17] are also well-studied. In this paper, we are concerned with a simpler phenomenon which occurs frequently in a military operation. In a battlefield, units that belong to different leagues are deployed in some bases of operation. Each base has to assure the supply line via controlling the communication channels, like bridges or roads adjacent to the base. Intuitively, a communication channel can be controlled by the unit which deploys more number of troops. Such a circumstance can be formulated as a strategic game, where players are the units, strategies are the deployments of troops, and the preferences are determined according to the number of communication channels controlled.

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The rest of this paper is organized as follows. In Section 2, a formal model will be given using a graph. Under the assumption that each unit has the same number troops R and that each communication channel can be captured using a bounded number of troops C , some properties of a Nash equilibrium are elaborated. For the graph being complete, a necessary condition are given in Section 3 for the case where $R = (n - 2)C$, and n is the number of units (players). In Section 4, experimental results are shown to demonstrate the constitution of the Nash equilibria for $R = kC$, $k \in \mathbb{N}$. Finally, concluding remarks are given in Section 5.

2 Model

Given is a graph $G = (V, E)$ and $R, C \in \mathbb{N}$. For convenience, we assume that $V = \{1, 2, \dots, n\}$. The *route-capturing game* is defined as a strategic game $\Gamma = (G, C, R)$, where

- V is the set of players,
- For each player i , a strategy s_i of i is a *feasible* n -tuple $(r_{i1}, r_{i2}, \dots, r_{in})$, where r_{ij} is an integer belonging to $[0, C]$, $r_{ij} > 0$ if $ij \in E$, and $\sum_{j=1}^n r_{ij} \leq R$.
- The preference of player i is defined according to the utility function u_i . For a strategy vector $s = (s_1, s_2, \dots, s_n)$, $u_i(s) = w_i - \ell_i$, where $w_i = |\{j : r_{ij} > r_{ji}\}|$ and $\ell_i = |\{j : r_{ij} < r_{ji}\}|$.

The utility function u_i may be defined equivalently as

$$u_i(s) = \sum_{j=1}^n \frac{r_{ij} - r_{ji}}{\max\{1, |r_{ij} - r_{ji}|\}},$$

which will be used alternatively later in this paper. For each player i , w_i , ℓ_i , and d_i are said to be the number of *wins*, *losses*, and *draws* of i , respectively, where $d_i = |\{j : r_{ij} = r_{ji}\}|$. We append

for each i a *score vector* (w_i, d_i, ℓ_i) . The following fact holds straightforwardly.

Fact 1 : For any strategy vector $s = (s_1, s_2, \dots, s_n)$, $\sum_{i \in V} w_i = \sum_{i \in V} \ell_i$.

Given a strategy vector s , we denote by s_{-i} the vector $(s_1, s_2, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$. A draw between i and j is said to be *non-full* if $r_{ij} = r_{ji} < C$, and the number of non-full draws is denoted by $d'_i = |\{j : r_{ij} = r_{ji} \text{ and } r_{ij} < C\}|$. A strategy s_i is said to be *binary* if $r_{ij} \in \{0, 1\}, \forall j \in V$, and *full-binary* if $r_{ij} \in \{0, C\}, \forall j \in V$, where $C > 1$. Analogously, a strategy vector s is binary and full-binary if s consists of binary and full-binary strategies, respectively.

Some properties the route-capturing game are given below.

Lemma 1 : If s is a binary strategy vector and $\sum_{j=1}^n r_{ij} = \min\{R, \deg(i) \cdot C\}, \forall i \in V$, then s is a Nash equilibrium.

Proof: For any $i \in V$, let $s'_i = (r'_{i1}, r'_{i2}, \dots, r'_{in})$ be an arbitrary strategy of i . Since s is binary, we have that

$$r'_{ij} = 0 \text{ and } r_{ji} = 1, \text{ if } r'_{ij} < r_{ji}. \quad (1)$$

It follows that

$$\begin{aligned} & u_i(s'_i, s_{-i}) \\ &= \sum_{j=1}^n \frac{r'_{ij} - r_{ji}}{\max\{1, |r'_{ij} - r_{ji}|\}} \\ &\leq \sum_{r'_{ij} \geq r_{ji}} (r'_{ij} - r_{ji}) + \sum_{r'_{ij} < r_{ji}} \frac{r'_{ij} - r_{ji}}{\max\{1, |r'_{ij} - r_{ji}|\}} \\ &\stackrel{(1)}{=} \sum_{r'_{ij} \geq r_{ji}} (r'_{ij} - r_{ji}) + \sum_{r'_{ij} < r_{ji}} (r'_{ij} - r_{ji}) \\ &= \sum_{j=1}^n (r'_{ij} - r_{ji}) \\ &\leq \sum_{j=1}^n (r_{ij} - r_{ji}) \\ &= u_i(s_i, s_{-i}). \end{aligned}$$

The last inequality holds because $\sum_{j=1}^n r_{ij} = \min\{R, \deg(i) \cdot C\}$. Thus, s is a Nash equilibrium. $\square$

Lemma 2 : Let s^* be a Nash equilibrium of a route-capturing game Γ , and let $D_i = \{r_{ij} : r_{ij} = r_{ji}\}$ for $i \in V$. If $D_i \neq \emptyset$ and $\exists j, r_{ji}^* > r_{ij}^*$, then $r_{ji}^* \geq \max_{r \in D_i} r$.

Proof: Suppose to the contrary that $r_{ji}^* < r_{ik}^* \in D_i$. Let s_i be a strategy of i satisfying $r_{ij'} = r_{ij'}^*, \forall j' \in V \setminus \{j, k\}$, $r_{ij} = r_{ik}^*$, and $r_{ik} = r_{ij}^*$. We claim that $u_i(s^*) < u_i(s_i, s_{-i}^*)$. Since

$$\frac{r_{ij}^* - r_{ji}^*}{\max\{1, |r_{ij}^* - r_{ji}^*|\}} + \frac{r_{ik}^* - r_{ki}^*}{\max\{1, |r_{ik}^* - r_{ki}^*|\}} = -1 + 0$$

and

$$\frac{r_{ik}^* - r_{ji}^*}{\max\{1, |r_{ik}^* - r_{ji}^*|\}} + \frac{r_{ij}^* - r_{ki}^*}{\max\{1, |r_{ik}^* - r_{ki}^*|\}} = 1 + (-1),$$

it follows that $u_i(s^*) < u_i(s'_i, s_{-i}^*)$, which contradicts that s^* is a Nash equilibrium. $\square$

Lemma 3 : Let s^* be a Nash equilibrium of a route-capturing game Γ with $d'_i \geq 1$ for some $i \in V$. The following properties hold.

- If $\exists j$ such that $r_{ij} < r_{ji}$, then $r_{ij} = 0$.
- If $\exists j$ such that $r_{ji} < r_{ij}$, then $r_{ij} = r_{ji} + 1$.

Proof: Let $r_{ik} = r_{ki} < C$. If $\exists j, r_{ij} < r_{ji}$ and $r_{ij} > 0$, one may replace r_{ij} and r_{ik} with $r_{ij} - 1$ and $r_{ik} + 1$, respectively, to increase w_i without increasing ℓ_i . Similarly, if $\exists j, r_{ji} < r_{ij}$ and $r_{ij} > r_{ji} + 1$, u_i can be increased by replacing r_{ij} and r_{ik} with $r_{ij} - 1$ and $r_{ik} + 1$, respectively. $\square$

Lemma 4 : Let s^* be a Nash equilibrium of a route-capturing game (G, C, R) . For $i \in V$, if $d_i \geq 3$ and $\exists j, r_{ij} = r_{ji} \geq 2$, then $d'_i \leq 2$.

Proof: Let $j_1, j_2 \in V$ such that $r_{ij_1} = r_{j_1i} < C$ and $r_{ij_2} = r_{j_2i} < C$. Consider the strategy s_i of i by replacing r_{ij_1} , r_{ij_2} , and r_{ij} with $r_{ij_1} + 1$, $r_{ij_2} + 1$, and $r_{ij} - 2$, respectively. The number of wins becomes $w_i + 2$ and that of losses becomes $\ell_i - 1$. Thus, $u_i(s^*) = w_i - \ell_i < (w_i + 2) - (\ell_i + 1) = u_i(s_i, s_{-i}^*)$, which contradicts that s^* is a Nash equilibrium. $\square$

3 Nash equilibria of $(K_n, C, (n-2)C)$

In this section, we investigate the case where $G = K_n$. A straightforward observation can be derived immediately as in Proposition 1.

Proposition 1 : For $k \geq n - 1$, the route-capturing game (K_n, C, kC) contains exactly one Nash equilibrium s^* , which is a full strategy vector.

For $k \leq n - 2$, a Nash equilibrium may not be a full strategy vector. In the following, we give a necessary condition of a Nash equilibrium for the case where $k = n - 2$.

Lemma 5 : Let s^* be a Nash equilibrium of a route-capturing game (K_n, C, kC) for $k \in \mathbb{N}$, $n > 3$. For any $i \in V$, we have that $\ell_i \leq n - k - 1$.

Proof: Suppose to the contrary that $\ell_i \geq n - k$. Without loss of generality, we assume that $i = n$ and $r_{ij}^* < r_{ji}^*$ for $1 \leq j \leq \ell_i$. We claim that $u_i(s^*) < u_i(s_i, s_{-i}^*)$, where $s_i = (r_{i1}, r_{i2}, \dots, r_{in})$ and $r_{ij} = C$ for $n - k \leq j < n$. Apparently, $r_{in-k} \geq r_{n-k,i}^*$, and $r_{ij}^* > r_{ji}^*$ implies $r_{ij} > r_{ji}^*$ for $n - k < j < n$. Therefore, we have that

$$w_i^* \leq w_i, \quad (2)$$

where w_i^* and w_i are the number of wins of i w.r.t. s_i^* and (s_i, s_{-i}^*) , respectively. Let ℓ_i^* be the number of losses of i w.r.t. s_i^* , and let ℓ_i be that w.r.t. (s_i, s_{-i}^*) . It follows that

$$\ell_i = \ell_i^* - 1. \quad (3)$$

Thus,

$$\begin{aligned} u_i(s^*) &= w_i^* - \ell_i^* \\ &< w_i^* - (\ell_i^* - 1) \\ &\leq w_i - \ell_i \\ &\stackrel{(2),(3)}{=} u_i(s_i, s_{-i}^*). \end{aligned}$$

□

Lemma 6 : For any strategy vector s , if $\exists i$ such that $r_{ij_1} = r_{ij_2} = C > 1$, $r_{j_1 i} = r_{j_2 i} = 0$, and $r_{j_3 i} > r_{ij_3} = 0$, then s is not a Nash equilibrium.

Proof: Consider the strategy $s'_i = (r'_{i1}, r'_{i2}, \dots, r'_{in})$, where

$$r'_{ij} = \begin{cases} \lfloor \frac{r_{ij}}{2} \rfloor, & \text{if } j = j_1; \\ \lceil \frac{r_{ij}}{2} \rceil, & \text{if } j = j_2; \\ C, & \text{if } j = j_3; \\ r_{ij}, & \text{o.w.} \end{cases}$$

One can easily verify that s'_i is feasible and the number of losses is decreased by one. Thus, $u_i(s) < u_i(s'_i, s_{-i})$. □

Theorem 1 : In a route-capturing game $\Gamma = (K_n, C, (n - 2)C)$, Γ has a Nash equilibrium s^* . Moreover, either of the following properties holds:

- s^* is a binary strategy vector;
- s^* is a full-binary strategy vector, and $(w_i, d_i, \ell_i) \in \{(0, n - 1, 0), (1, n - 3, 1)\}, \forall i \in V$.

Proof: By Lemma 5, $\ell_i \leq 1, \forall i \in V$. For any player i , we consider the following cases for (w_i, d_i, ℓ_i) . We show in cases 1–4 the forbidden cases for the score vector, and then in cases 5–6 the possible strategies for each player.

- **Case 1:** $(n - 1, 0, 0)$

$\forall j, k \in V \setminus \{i\}$, $r_{jk} = r_{kj} = C$ since o.w. $\exists k, k'$ s.t. $0 < r_{jk} = r_{kj} < C$ and $0 < r_{jk'} < C$. $r'_{jk} = \min\{C, r_{jk} + r_{jk'}\} \dots$

By Lemma 2, $r_{ij} = C, \forall j \in V \setminus \{i\}$. Thus, $\sum_j r_{ij} = (n - 1)C > (n - 2)C$, a contradiction.

- **Case 2:** $(0, n - 2, 1)$:

Let i be a vertex with $(w_i, d_i, \ell_i) = (0, n - 2, 1)$, and let i' be the player satisfying $r_{i'i} > r_{ii'}$. It follows that $r_{ii'} = 0$ since otherwise $\exists k$ such that $r_{ik} = r_{ki} < C$, and by Lemma 3 s is not a Nash equilibrium. One can further derive that

$$r_{ij} = C, \forall j \neq i'. \quad (4)$$

claim 1: For any player $j \in V \setminus \{i\}$, if $r_{jk} > r_{kj}$ for some $k \neq j$, then $r_{jk} = C$ and $r_{kj} = 0$. Consider the player i' . Since $r_{ij} = r_{ji} = C$, by Lemma 2, it follows that

$$\forall j \neq i', r_{i'j} = C \text{ if } r_{i'j} > r_{ji'}. \quad (5)$$

Together with Lemma 3, we have that

$$r_{i'j} = C \text{ if } r_{i'j} = r_{ji'}. \quad (6)$$

By (5) and (6), we have that $\ell_{i'} = 1$ and

$$r_{i'j} = 0 \text{ if } r_{i'j} < r_{ji'}. \quad (7)$$

By Lemma 6, we have that $w_{i'} < 2$, and therefore

$$d_{i'} \geq 1. \quad (8)$$

For $k \neq i'$, by (6) and (8), Lemma 2 implies that

$$r_{kj} = C \text{ if } r_{kj} > r_{jk}. \quad (9)$$

Moreover,

$$r_{kj} = 0 \text{ if } r_{kj} < r_{jk}, \quad (10)$$

since otherwise $\exists j'$ such that $r_{kj'} = r_{j'k} < C$, and by Lemma 3 s is not a Nash equilibrium. To sum up, by (5), (7), (9), and (10), claim 1 follows.

claim 2: $(w_j, d_j, \ell_j) = (1, n-3, 1)$ for $j \in V_1 \setminus \{i\}$.

For $\ell_j = 1$, $w_j \leq 1$ since otherwise by Lemma 6, s is not a Nash equilibrium.

claim 3: $(w_j, d_j, \ell_j) = (0, n-1, 0)$ for $j \in V_0$.

It can be derive from (9) that if $w_j \geq 1$, $\exists k$ such that $r_{jk} = r_{kj} < C$, which, by Lemma 3, contradicts that s is a Nash equilibrium.

According to claim 2 and claim 3, Fact 1 leads to the following contradiction.

$$\begin{aligned} \sum_{i \in V} \ell_i &= \sum_{i \in V} w_i \\ &= \sum_{i \in V_0} w_i + \sum_{i \in V_1} w_i \\ &= |V_1| - 1 \\ &< |V_1| = \sum_{i \in V} \ell_i. \end{aligned}$$

- **Case 3:** $(n-2, 0, 1)$

For $j \in W_i$, by Lemma 5, $j \in D_{j'}$ for $j' \in W_i \setminus \{j\}$. For $k \in L_i$, since $n \geq 4$, Lemma 5 implies that $\exists j^* \in W_i$, $j^* \in D_k$. Thus, we have that $(w_{j^*}, d_{j^*}, \ell_{j^*}) = (0, n-2, 1)$, which is a contradiction.

- **Case 4:** $(x, n-1-x, 0)$, $1 \leq x < n-1$

For $j, j' \in W_i$, $r_{jj'} = r_{j'j}$ since otherwise either $\ell_j \geq 2$ or $\ell_{j'} \geq 2$, contradicts Lemma 5. Moreover, $r_{jk} > r_{kj}$ for some $k \in D_i$ since otherwise $d_j = n-2$. Let $V_0 = \{i\}$, by repeating the argument above, V has to be able to be partitioned as $\bigcup_j V_j$ such that $r_{jk} < r_{kj}$ and $r_{jk'} > r_{k'j}$ for $j \in V_i$, $k \in V_{i-1}$, and $k' \in V_{i+1}$. However, since $|V|$ is finite, such a partition does not exist.

- **Case 5:** $(0, n-1, 0)$:

We claim that $r_{ij} = 0$ or $r_{ij} = C, \forall i, j \in V$. Suppose to the contrary that $\exists j, 0 < r_{ij} < C$. Since $R = (n-2)C$ and $d_i = n-1$, by the pigeonhole principle, $|\{j : r_{ij} = r_{ji} < C\}| \geq 2$. Moreover, since $n \geq 4$, by Lemma 4, there is a contradiction.

- **Case 6:** $(x, n-2-x, 1)$, where $0 < x < n-1$

According to cases 1–4, V can be partitioned as $V_0 = \{j : w_j = \ell_j = 0\}$ and $V_1 = \{j : w_j \geq 1 \text{ and } \ell_j = 1\}$. By Fact 1, we have that

$$|V_1| \leq \sum_{j \in V_1} w_j = \sum_{j \in V} w_j = \sum_{j \in V} \ell_j = \sum_{j \in V_1} \ell_j = |V_1|,$$

which implies that $w_j = 1, \forall j \in V_1$.

We claim that $r_{ij} = 0$ if $r_{ij} < r_{ji}$. Suppose to the contrary that $r_{ij} > 0$, by Lemma 3, $r_{ij'} = r_{j'i} = C$. Thus, $r_{ik} < C$ for $r_{ik} > r_{ki}$. By Lemma 2, we have that $r_{kj''} \leq r_{ik} < C, \forall r_{kj''} = r_{j''k}$. Again, by Lemma 3, $r_{kj''} = 0$ if $r_{kj''} < r_{j''k}$. Thus, $\sum_j r_{kj} < (n-2)C$, which contradicts that s is a Nash equilibrium, and the claim follows. One can further derive that $r_{ij} = C, \forall r_{ij} \geq r_{ji}$.

To sum up, $(w_i, d_i, \ell_i) \in \{(0, n-2, 0), (1, n-3, 1)\}, \forall i \in V$, and s_i is full-binary. Together with Lemma 1, the theorem follows. $\square$

4 Experimental results

In this section, experimental results are demonstrated to elaborate the properties of all the Nash equilibria of a route-capturing game (K_n, C, kC) . The idea is to enumerate all possible strategy vectors and then verify which of those are the Nash equilibria. For a given strategy vector s , the verification is done by computing a *best response* s'_i for each player i w.r.t. s_{-i} , i.e., $u_i(s'_i, s_{-i}) \geq u_i(s''_i, s_{-i})$ for any strategy s''_i , and then compare $u_i(s'_i, s_{-i})$ to $u_i(s)$.

Given a strategy vector s , a best response s'_i of i w.r.t. s_{-i} can be computed by applying a greedy approach, which depends primarily on the following lemma.

Lemma 7: Let s be a strategy vector. There is a best response $s'_i = (r'_{i1}, r'_{i2}, \dots, r'_{in})$ of i w.r.t. s_{-i} such that $r'_{ij} > r_{ji} = \min_j \{r_{ji} : j \in E\}$.

Let r_{ji} be sorted in ascending order as $r_{j_0i}, r_{j_1i}, \dots, r_{j_{n-1}i}$. Without loss of generality, we assume that $j_0 = i$. We define $\tilde{s}_i = (\tilde{r}_{i1}, \tilde{r}_{i2}, \dots, \tilde{r}_{in})$ as

$$\tilde{r}_{ijk} = \begin{cases} 0, & \text{if } k = 0; \\ \eta_k, & \text{if } \xi_k > \eta_k; \\ \xi_k, & \text{o.w.,} \end{cases}$$

Table 1: Experimental results of (K_4, C, kC)

(C, k)	#01	#0C	OTHERS	STRATEGIES
(1, 1)	81	0	0	81
(2, 1)	81	0	0	1,296
(3, 1)	1	0	0	10,000
(4, 1)	0	0	0	50,625
(5, 1)	0	0	0	194,481
(6, 1)	0	0	0	614,656

Table 2: Experimental results of (K_5, C, kC)

(C, k)	#01	#0C	OTHERS	STRATEGIES
(1, 1)	1,024	0	0	1,024
(1, 2)	7,776	0	0	7,776
(1, 3)	1,024	0	0	1,024
(2, 1)	7,776	0	0	100,000
(2, 2)	1	0	0	2,476,099
(2, 3)	0	44	0	100000
(3, 1)	1,024	0	0	3,200,000
(3, 2)	0	0	0	164,916,224
(3, 3)	0	44	0	3,200,000

where $\xi_k = R - \sum_{l=1}^{k-1} r'_{ijl}$ and $\eta_k = \min\{r_{jki} + 1, C\}$.

Lemma 8: Given a strategy vector s , the strategy $\tilde{s}_i$ is a best response of i w.r.t. s_{-i} .

By Lemma 8, for a given strategy vector s , one may determine if s is a Nash equilibrium in polynomial time. Experimental results for (K_4, C, kC) , (K_5, C, kC) , and (K_6, C, kC) are shown in Tables 1–3. Specifically, for (K_4, C, kC) , we test only the case where $k = 1$ since for $k \geq 2$, the Nash equilibria of (K_4, C, kC) are characterized by Proposition 1 and Theorem 1.

Each row of the tables reveals the distribution of the Nash equilibria of (K_n, C, kC) , where C and k are set according to each cell of column “ (C, k) ”. The cells of columns #01 and #0C record the number of binary and full-binary Nash equilibria in the corresponding game, and the number of all the other Nash equilibria are recorded in the cell of column OTHERS. For each (K_n, C, kC) , the number of strategy vectors are recorded in column STRATEGIES.

As observed from Table 1, we have the following conjecture.

Conjecture 2: In a route-capturing game $\Gamma = (K_4, C, C)$, Γ has a Nash equilibrium s^* iff $C \leq 3$. Moreover, s^* is binary.

A more general conjecture can be made by observing Tables 2 and 3. As one can see, a Nash equilibrium exists when $k \geq n - 2$ (Proposition 1 and Theorem 1) and $kC \leq n - 1$ (Lemma 1). This leads us to give the following conjecture, which is a necessary and sufficient condition of the existence of a Nash equilibrium.

Conjecture 3 : A route-capturing game (K_n, C, kC) contains a Nash equilibrium s^* iff $k \geq n - 2$ or $kC \leq n - 1$. Moreover, s^* is binary if $kC \leq n - 1$, and full-binary if $k \geq n - 2$.

5 Concluding remarks

In this paper, we are concerned with the route-capturing game (G, C, R) . Some properties are elaborated for G being a general graph. Specifically, we give a necessary condition for a Nash equilibrium of (K_n, C, kC) . Some conjectures are made according to the experimental results. Whether the conjectures hold or not will be conducted as future works.

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Table 3: Experimental results of (K_6, C, kC)

(C, k)	#01	#0C	OTHERS	STRATEGIES
(1, 1)	15,625	15,625	0	15,625
(1, 2)	1,000,000	1,000,000	0	1,000,000
(1, 3)	1,000,000	1,000,000	0	1,000,000
(1, 4)	15,625	15,625	0	15,625
(2, 1)	1,000,000	0	0	11,390,625
(2, 2)	15625	0	0	8,303,765,625
(2, 3)	0	0	0	8,303,765,625
(2, 4)	0	265	0	11,390,625

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